IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

SOFTWARE SYSTEMS

Thursday, 6 May 10:00 am

Time allowed: hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

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SOFTWARE SYSTEMS

1. a) Invent an example of a composite attribute you might find in a real-world data model, giving both the composite name and the sub-parts. You cannot use addresses or names as your example. [5]

   b) Invent an example of a foreign-key relationship in a real-world data model. You cannot use relationships involving people as your example. [5]

   c) How many rows and columns will the query `SELECT * FROM orders WHERE ProductId = 'xt23341';` produce? [5]

   d) What SQL operator is closest to the $\rtimes$ operator, and what would $A \rtimes B$ return if B contains no rows? [5]

   e) Translate these two C structures into equivalent relations. [5]

```
struct Company{
  string name;
};
struct Component{

  string key;
  // 0 or more companies may make a component
  vector<Company *> manufacturers;
}
```

   f) State whether the following statement is true or false, and briefly explain why: "HTTP can only work over Ethernet networks." [5]

   g) If a computer does not have access to a DNS server, can it still send and receive IP packets? Explain your answer. [5]

   h) Give one theoretical advantage of a packet-switched network over a circuit-switched network, and give a concrete example of where that advantage matters in practice. [5]

   i) What is special about IP port numbers less than 1024, and give an example of a piece of software which binds to a port below 1024. [5]

   j) What is the purpose of the checksum field in an IP packet? [5]

2. You have been asked to design the network and data architecture for a system containing very large numbers of low-power sensor nodes. Each sensor node is able to determine both its position and the current time using GPS, and can measure the local temperature, pressure, and humidity through on-board sensors. A major constraint on the nodes is battery life – the nodes have a small solar panel and battery, and due to power constraints must spend a lot of time sleeping.

   The sensor nodes contain a wireless network interface which allow them to send and receive packets over long distances, though this can only happen infrequently due to the power constraints. All nodes are within communication range of at least one server, each of which is a standard server with a mains power supply. All servers know their own GPS location, and can communicate with each other using a wired network.

   a) Each server has no initial knowledge about the number and positions of nodes and other servers, and must build a model of them as it receives sensor readings from them.

      i) Identify three entity sets within this system. [3]

      ii) Identify a natural primary key within the system, stating any assumptions you make. [3]

      iii) Draw an ER diagram of the system containing your three entity sets, and at least one relationship set involving each entity set. [6]

      iv) Describe a many-to-many relationship set in the system. This may or may not be a relationship set shown in your ER diagram. [3]

   b) The servers are connected via a wired network, but do not initially know how many other servers there are or their addresses.

      i) Assume the servers are connected together by a single Ethernet switch, but are not connected to the global internet. How could the servers find each other? [4]

      ii) Now assume the servers are on different LANs, but are all connected to the global internet. How could the servers find each other? [4]

c) By default the nodes try to report their latest sensor readings every hour on the hour, using the GPS clock to work out when to send, and then sending a single packet using a connection-less protocol. This works fairly well for small test deployments using $n = 100$ nodes, but when deployed with $n = 1000$ nodes many updates are lost, and for $n = 10000$ nodes very few update packets get through due to packet loss.

i) Explain the problem that is occurring. [3]

ii) Develop a simple mathematical model to estimate the rate of packet loss as a function of the number of nodes. Describe any extra parameters needed and any assumptions made. This model does not need to be exact, but should quantitatively explain why the problem becomes substantially worse as n increases. [5]

iii) If the goal is simply to receive one sensor reading per hour *on average*, and the sensor node energy budget only allows the sending of one packet in each hour, propose a change to the sending strategy that increases the chances of each sensors hourly update packet being received. [4]

iv) The goal is now to try to capture all sensor readings as close to the hour as possible, but the energy budget does allow more than one packet to be sent per hour. Propose a new sending strategy for this use-case. [4]

d) Assume that the data received from the sensors has been post-processed and aggregated into distinct regions, and exported to two tables with the following structure:

```
CREATE TABLE regions (
    id STRING PRIMARY KEY,
    x FLOAT,
    y FLOAT
);
CREATE TABLE daily_samples (
    region_id STRING FOREIGN KEY REFERENCES regions(id)
    date DATE,
    temperature FLOAT,
    pressure FLOAT,
    humidity FLOAT
);
```

Note: these tables are not directly related to the data model described earlier, and would not be expected to appear in the ER model.

i) Give an SQL query that counts the number of regions. [3]

ii) Give an SQL query that returns the average temperature on a given date across all regions. [4]

iii) If there are N regions and M readings and every reading refers to a valid region, what is the maximum number of rows if we join the two tables on the region id? [4]